

Mezinárodní programy vysokoškolských kurzů matematiky

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Integritní omezení (IO)

UNIVERZITY: univerzitní ID (unikátní), název (unikátní), stát (USA, UK, France, Czech republic, Germany, Spain, China; necht' každá univerzita se nachází ve právě jedním státu).

- Osobní ID musí být mezi 100 a 999 včetně.
- Může obsahovat 1-N programů v různých jazycích.

Každá univerzita má několik PROGRAMŮ:

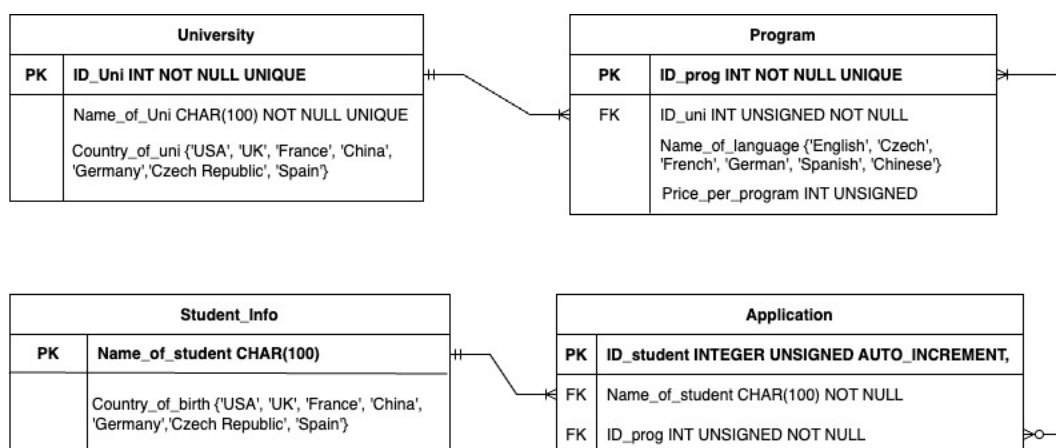
- ID programu (unikátní) musí být mezi 10000 a 99999 včetně.
- Program zařizuje konkrétní univerzita.
- Jazyk programu je English, Czech, Chinese, French, German, Spanish.
- Cena programu za rok v dolarech.
- Může obsahovat 1-N studentů (jinak program musí být vyčten z databáze).

Každý student podal ŽÁDOST: pořádkové číslo žádosti (unikátní), jméno studentu, ID programu.

STUDENTI, kteří jsou zapsaní do programu(ů): jméno studentu (unikátní), občanství – místo narození (USA, UK, France, Czech republic, Germany, Spain, China).

- Student může mít 1-N žádosti (jinak program musí být vyčten z databáze).

ERA-Model



SQL

Realizováno v databázovém systému MariaDB verze 10.5 (na školním serveru LX).

Tabulky (včetně cizích klíčů a dalších omezení)

```
ALTER DATABASE ermakece_ZAPOCET_ERMAKECA CHARACTER SET utf8mb4 COLLATE  
utf8mb4_czech_ci;
```

```
CREATE TABLE University (  
    ID_uni INTEGER UNSIGNED NOT NULL UNIQUE,  
    Name_of_uni VARCHAR (100) NOT NULL UNIQUE,  
    Country_of_uni ENUM ('USA', 'UK', 'France', 'China', 'Germany', 'Czech Republic',  
    'Spain') NOT NULL,  
    PRIMARY KEY (ID_uni)  
);
```

```
CREATE TABLE Program (  
    ID_prog INTEGER UNSIGNED NOT NULL UNIQUE,  
    ID_uni INTEGER UNSIGNED NOT NULL,  
    Name_of_language ENUM ('English', 'Czech', 'French', 'German', 'Spanish', 'Chinese')  
    NOT NULL,  
    Price_per_program INTEGER UNSIGNED,  
    PRIMARY KEY (ID_prog),  
    FOREIGN KEY (ID_uni) REFERENCES University (ID_uni) ON DELETE CASCADE  
);
```

```
CREATE TABLE Student_Info (  
    Name_of_student VARCHAR (100) NOT NULL,  
    Country_of_birth ENUM ('USA', 'UK', 'France', 'China', 'Germany', 'Czech Republic',  
    'Spain') NOT NULL,  
    UNIQUE (Name_of_student),  
    PRIMARY KEY (Name_of_student)  
);
```

```
CREATE TABLE Application (  

```

```
ID_Application INTEGER UNSIGNED AUTO_INCREMENT,  
Name_of_student VARCHAR (100) NOT NULL,  
ID_prog INTEGER UNSIGNED NOT NULL,  
PRIMARY KEY (ID_Application),  
FOREIGN KEY (ID_prog) REFERENCES Program (ID_prog) ON DELETE CASCADE  
FOREIGN KEY (Name_of_student) REFERENCES Student_Info (Name_of_student)  
ON DELETE CASCADE  
);
```

```
ALTER TABLE University ADD CONSTRAINT ID_U CHECK (ID_uni >= 100 AND ID_uni <  
1000);
```

```
ALTER TABLE Program ADD CONSTRAINT ID_P CHECK (ID_prog >= 10000 and ID_prog <  
100000);
```

Data

```
INSERT INTO University VALUES (101, 'Czech Technical University', 'Czech Republic');  
INSERT INTO University VALUES (102, 'Charles University', 'Czech Republic');  
INSERT INTO University VALUES (201, 'Harvard University', 'USA');  
INSERT INTO University VALUES (202, 'Californian Technical University', 'USA');  
INSERT INTO University VALUES (203, 'Massachusetts Institute of Technology', 'USA');  
INSERT INTO University VALUES (204, 'Stanford University', 'USA');  
INSERT INTO University VALUES (301, 'University of Cambridge', 'UK');  
INSERT INTO University VALUES (302, 'The University of Edinburgh', 'UK');  
INSERT INTO University VALUES (303, 'University of Manchester', 'UK');  
INSERT INTO University VALUES (401, 'Sorbonne University', 'France');  
INSERT INTO University VALUES (402, 'University PSL', 'France');  
INSERT INTO University VALUES (501, 'Technical University of Munich', 'Germany');  
INSERT INTO University VALUES (502, 'Technische Universität Berlin', 'Germany');  
INSERT INTO University VALUES (601, 'Complutense University of Madrid', 'Spain');  
INSERT INTO University VALUES (602, 'Universitat de Barcelona', 'Spain');  
INSERT INTO University VALUES (603, 'Universidad Polit cnica de Madrid', 'Spain');
```

```
INSERT INTO University VALUES (701, 'Fudan University', 'China');
INSERT INTO University VALUES (702, 'Shanghai Jiao Tong University', 'China');

INSERT INTO Program VALUES (10011, 101, 'Czech', 0);
INSERT INTO Program VALUES (10012, 101, 'English', 2000);
INSERT INTO Program VALUES (10021, 102, 'Czech', 0);
INSERT INTO Program VALUES (10022, 102, 'English', 3000);

INSERT INTO Program VALUES (20011, 201, 'English', 60000);
INSERT INTO Program VALUES (20021, 202, 'English', 30000);
INSERT INTO Program VALUES (20031, 203, 'English', 40000);
INSERT INTO Program VALUES (20041, 204, 'English', 45000);
INSERT INTO Program VALUES (30011, 301, 'English', 30000);
INSERT INTO Program VALUES (30021, 302, 'English', 20000);
INSERT INTO Program VALUES (30031, 303, 'English', 15000);
INSERT INTO Program VALUES (40011, 401, 'French', 16000);
INSERT INTO Program VALUES (40021, 402, 'French', 10000);
INSERT INTO Program VALUES (40012, 401, 'English', 32000);
INSERT INTO Program VALUES (40022, 402, 'English', 20000);
INSERT INTO Program VALUES (50011, 501, 'German', 3000);
INSERT INTO Program VALUES (50021, 502, 'German', 3000);
INSERT INTO Program VALUES (50012, 501, 'English', 6000);
INSERT INTO Program VALUES (50022, 502, 'English', 6500);
INSERT INTO Program VALUES (60011, 601, 'Spanish', 2000);
INSERT INTO Program VALUES (60021, 602, 'Spanish', 3000);
INSERT INTO Program VALUES (60031, 603, 'Spanish', 1000);
INSERT INTO Program VALUES (60012, 601, 'English', 4000);
INSERT INTO Program VALUES (60022, 602, 'English', 5500);
INSERT INTO Program VALUES (60032, 603, 'English', 2500);
INSERT INTO Program VALUES (70012, 701, 'English', 8000);
```

INSERT INTO Program VALUES (70022, 702, 'English', 6000);

INSERT INTO Program VALUES (70011, 701, 'Chinese', 0);

INSERT INTO Program VALUES (70021, 702, 'Chinese', 3500);

INSERT INTO Student_Info VALUES ('Ahmed Al-Sayed', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Aisha Khan', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Alejandro Navarro', 'China');

INSERT INTO Student_Info VALUES ('Anastasia Morozova ', 'Spain');

INSERT INTO Student_Info VALUES ('Anna Müller', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Camila Carvalho', 'France');

INSERT INTO Student_Info VALUES ('Diego Castillo', 'Spain');

INSERT INTO Student_Info VALUES ('Elias Haddad', 'France');

INSERT INTO Student_Info VALUES ('Emilia Garcia ', 'USA');

INSERT INTO Student_Info VALUES ('Fatima Ahmed', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Grace Johnson ', 'Spain');

INSERT INTO Student_Info VALUES ('Hana Jimkov ', 'Spain') ;

INSERT INTO Student_Info VALUES ('Haruto Sato', 'France');

INSERT INTO Student_Info VALUES ('Hugo Sanchez ', 'Spain');

INSERT INTO Student_Info VALUES ('Charlotte Lefevre ', 'UK');

INSERT INTO Student_Info VALUES ('Ibrahim Diallo', 'UK');

INSERT INTO Student_Info VALUES ('Ingrid Johansen', 'China');

INSERT INTO Student_Info VALUES ('Ivan Petrovic', 'Germany');

INSERT INTO Student_Info VALUES ('Jae-Hyun Kim', 'UK');

INSERT INTO Student_Info VALUES ('Javier Ortega', 'USA');

INSERT INTO Student_Info VALUES ('Jorrit van Dijk ', 'Spain');

INSERT INTO Student_Info VALUES ('Julia Kowalska', 'USA');

INSERT INTO Student_Info VALUES ('Laura Jones', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Leonardo Almeida', 'Germany');

INSERT INTO Student_Info VALUES ('Liam O Conno', 'Czech Republic');

INSERT INTO Student_Info VALUES ('Linh Nguyen ', 'Spain');

```
INSERT INTO Student_Info VALUES ('Lucas da Silva', 'USA');
INSERT INTO Student_Info VALUES ('Lucia Morales', 'China');
INSERT INTO Student_Info VALUES ('Marco Romano', 'Germany');
INSERT INTO Student_Info VALUES ('Maya Al-Mansouri', 'UK');
INSERT INTO Student_Info VALUES ('Miguel Torres', 'UK');
INSERT INTO Student_Info VALUES ('Natalia D az ', 'France');
INSERT INTO Student_Info VALUES ('Nikos Papadopoulos', 'USA');
INSERT INTO Student_Info VALUES ('Noor El-Sheikh', 'Germany');
INSERT INTO Student_Info VALUES ('Nora Becker ', 'Spain');
INSERT INTO Student_Info VALUES ('Nur Aini', 'Spain');
INSERT INTO Student_Info VALUES ('Pavel Novak', 'Spain');
INSERT INTO Student_Info VALUES ('Rahul Sharma', 'USA');
INSERT INTO Student_Info VALUES ('Ranjit Singh', 'France');
INSERT INTO Student_Info VALUES ('Selina Meier ', 'Germany');
INSERT INTO Student_Info VALUES ('Siti Nurhaliza', 'Germany');
INSERT INTO Student_Info VALUES ('Sofia Rossi', 'Czech Republic');
INSERT INTO Student_Info VALUES ('Sophia Andersson', 'China');
INSERT INTO Student_Info VALUES ('Sven Olsson', 'France');
INSERT INTO Student_Info VALUES ('Takumi Yamada', 'China');
INSERT INTO Student_Info VALUES ('Valentina Bianchi', 'China');
INSERT INTO Student_Info VALUES ('Victoria Petrescu', 'UK');
INSERT INTO Student_Info VALUES ('Viktorija Jankovic', 'Spain');
INSERT INTO Student_Info VALUES ('Yara Farouk', 'France');
INSERT INTO Student_Info VALUES ('Yuki Fujimoto', 'Spain');
INSERT INTO Student_Info VALUES ('Zoe Brown', 'France');
```

```
INSERT INTO Application VALUES (NULL, 'Laura Jones', 10011);
INSERT INTO Application VALUES (NULL, 'Aisha Khan', 10011);
INSERT INTO Application VALUES (NULL, 'Fatima Ahmed', 10012);
INSERT INTO Application VALUES (NULL, 'Anna Müller', 10012);
```

```
INSERT INTO Application VALUES (NULL, 'Liam O Conno', 10021);
INSERT INTO Application VALUES (NULL, 'Sofia Rossi', 10022);
INSERT INTO Application VALUES (NULL, 'Ahmed Al-Sayed', 10021);
INSERT INTO Application VALUES (NULL, 'Rahul Sharma', 20011);
INSERT INTO Application VALUES (NULL, 'Julia Kowalska', 20031);
INSERT INTO Application VALUES (NULL, 'Lucas da Silva', 20031);
INSERT INTO Application VALUES (NULL, 'Javier Ortega', 20041);
INSERT INTO Application VALUES (NULL, 'Emilia Garcia', 20041);
INSERT INTO Application VALUES (NULL, 'Nikos Papadopoulos', 20021);
INSERT INTO Application VALUES (NULL, 'Victoria Petrescu', 30011);
INSERT INTO Application VALUES (NULL, 'Jae-Hyun Kim', 30011);
INSERT INTO Application VALUES (NULL, 'Ibrahim Diallo', 30021);
INSERT INTO Application VALUES (NULL, 'Charlotte Lefevre', 30031);
INSERT INTO Application VALUES (NULL, 'Miguel Torres', 30031);
INSERT INTO Application VALUES (NULL, 'Maya Al-Mansouri', 30011);
INSERT INTO Application VALUES (NULL, 'Yara Farouk', 40011);
INSERT INTO Application VALUES (NULL, 'Haruto Sato', 40011);
INSERT INTO Application VALUES (NULL, 'Camila Carvalho', 40012);
INSERT INTO Application VALUES (NULL, 'Zoe Brown', 40012);
INSERT INTO Application VALUES (NULL, 'Ranjit Singh', 40022);
INSERT INTO Application VALUES (NULL, 'Sven Olsson', 40021);
INSERT INTO Application VALUES (NULL, 'Natalia Diaz', 40022);
INSERT INTO Application VALUES (NULL, 'Elias Haddad', 40021);
INSERT INTO Application VALUES (NULL, 'Selina Meier', 50011);
INSERT INTO Application VALUES (NULL, 'Leonardo Almeida', 50011);
INSERT INTO Application VALUES (NULL, 'Siti Nurhaliza', 50012);
INSERT INTO Application VALUES (NULL, 'Ivan Petrovic', 50012);
INSERT INTO Application VALUES (NULL, 'Noor El-Sheikh', 50021);
INSERT INTO Application VALUES (NULL, 'Marco Romano', 50022);
INSERT INTO Application VALUES (NULL, 'Sophia Andersson', 70022);
```

```
INSERT INTO Application VALUES (NULL, 'Lucia Morales', 70022);
INSERT INTO Application VALUES (NULL, 'Alejandro Navarro', 70021);
INSERT INTO Application VALUES (NULL, 'Ingrid Johansen', 70011);
INSERT INTO Application VALUES (NULL, 'Valentina Bianchi', 70011);
INSERT INTO Application VALUES (NULL, 'Takumi Yamada', 70012);
INSERT INTO Application VALUES (NULL, 'Grace Johnson ', 60011);
INSERT INTO Application VALUES (NULL, 'Pavel Novak', 60011);
INSERT INTO Application VALUES (NULL, 'Nora Becker ', 60012);
INSERT INTO Application VALUES (NULL, 'Diego Castillo', 60012);
INSERT INTO Application VALUES (NULL, 'Anastasia Morozova ', 60021);
INSERT INTO Application VALUES (NULL, 'Viktorija Jankovic', 60021);
INSERT INTO Application VALUES (NULL, 'Jorrit van Dijk ', 60022);
INSERT INTO Application VALUES (NULL, 'Yuki Fujimoto', 60022);
INSERT INTO Application VALUES (NULL, 'Linh Nguyen ', 60031);
INSERT INTO Application VALUES (NULL, 'Nur Aini', 60031);
INSERT INTO Application VALUES (NULL, 'Hugo S nchez ', 60032);
INSERT INTO Application VALUES (NULL, 'Hana imkov ', 60032 );
INSERT INTO Application VALUES (NULL, 'Laura Jones', 20021);
INSERT INTO Application VALUES (NULL, 'Aisha Khan', 20031);
INSERT INTO Application VALUES (NULL, 'Fatima Ahmed', 40021);
INSERT INTO Application VALUES (NULL, 'Anna Müller', 50021);
INSERT INTO Application VALUES (NULL, 'Lucas da Silva', 60012);
INSERT INTO Application VALUES (NULL, 'Javier Ortega', 70012);
INSERT INTO Application VALUES (NULL, 'Emilia Garc a ', 50012);
INSERT INTO Application VALUES (NULL, 'Nikos Papadopoulos', 30031);
INSERT INTO Application VALUES (NULL, 'Miguel Torres', 40021);
INSERT INTO Application VALUES (NULL, 'Maya Al-Mansouri', 40022);
INSERT INTO Application VALUES (NULL, 'Sven Olsson', 50012);
INSERT INTO Application VALUES (NULL, 'Elias Haddad', 10012);
INSERT INTO Application VALUES (NULL, 'Selina Meier', 10022);
```



```
INSERT INTO Application VALUES (NULL, 'Valentina Bianchi', 50011);
```

```
INSERT INTO Application VALUES (NULL, 'Grace Johnson', 60011);
```

```
INSERT INTO Application VALUES (NULL, 'Nur Aini', 70022);
```

```
INSERT INTO Application VALUES (NULL, 'Hugo Sanchez ', 70012);
```

Dotazy

1. Vypis všech studentů, kteří jsou Američany a studují v Evropě:

```
SELECT Application.Name_of_student AS StudentName,  
University.Country_of_uni AS StudyCountry  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni = Program.ID_uni  
INNER JOIN Student_Info ON Application.Name_of_student = Student_Info.Name_of_student  
WHERE Student_Info.Country_of_birth='USA'  
AND University.Country_of_uni IN ('France', 'Germany', 'Czech Republic', 'Spain');
```

2. Vypis všech studentů, kteří studují v Číně, v abecedním pořádku:

```
SELECT Application.Name_of_student AS StudentName  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni= Program.ID_uni  
WHERE University.Country_of_uni='China'  
ORDER BY StudentName;
```

3. Univerzita + počet student zápsaných do programu v dané univerzitě:

```
SELECT COUNT(*) AS Number_of_students,  
University.Name_of_uni AS University  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni= Program.ID_uni  
GROUP BY University.ID_uni;
```

4. Studenti, kteří platí celkově za zápsané programy minimalně 2000\$/rok:

```
SELECT Application.Name_of_student AS StudentName,  
SUM(Program.Price_per_program) AS Total_payment  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni= Program.ID_uni  
GROUP BY Application.Name_of_student  
HAVING SUM(Program.Price_per_program)>=2000;
```

5. Výpis třech studentů, kteří platí největší částku za zápsané programy:

```
SELECT Application.Name_of_student AS StudentName,  
SUM(Program.Price_per_program) AS Total_payment  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni= Program.ID_uni  
GROUP BY Application.Name_of_student  
ORDER BY SUM(Program.Price_per_program) DESC  
LIMIT 3;
```

6. Výpis počtu programu, kteří se provádí v angličtině:

```
SELECT COUNT(*) AS Number_of_programs_in_english  
FROM Program  
WHERE Name_of_language = 'English';
```

7. Výpis univerzit, které mají programy zdarma + jazyk programu:

```
SELECT University.Name_of_uni AS Free_University,  
Program.Name_of_language AS Language  
FROM Program  
INNER JOIN University ON University.ID_uni= Program.ID_uni  
WHERE Program.Price_per_program=0;
```

8. Částka, kterou UK vydělává na studentech, kteří přijíždějí ze zahraničí:

```
SELECT SUM(Program.Price_per_program) AS UK_profit_from_abroad  
FROM Program  
INNER JOIN Application ON Application.ID_prog = Program.ID_prog  
INNER JOIN University ON University.ID_uni = Program.ID_uni
```

INNER JOIN Student_Info ON Application.Name_of_student = Student_Info.Name_of_student

WHERE University.Country_of_uni = 'UK' AND Student_Info.Country_of_birth != 'UK';

9. Počet studentů studujících v angličtině:

SELECT COUNT(DISTINCT Application.Name_of_student) AS Students_who_study_in_English

FROM Program

INNER JOIN Application ON Application.ID_prog = Program.ID_prog

WHERE Program.Name_of_language = 'English';

10. Výpis studentů, které mají programy zdarma + jazyk programu:

SELECT Application.Name_of_student AS Student_study_for_free,

Program.Name_of_language AS Language

FROM Program

INNER JOIN Application ON Application.ID_prog = Program.ID_prog

INNER JOIN University ON University.ID_uni= Program.ID_uni

WHERE Program.Price_per_program=0;